

WHAT SURVIVES THE FALL

Four surfaces, from forgetting to acting

BLOW A SOAP BUBBLE and it becomes a sphere. It does not matter who blew it, how hard, or what strange shape the film passed through on the way. **Surface tension** pulls the film into the smallest area that will wrap the trapped air, and once it gets there it stops. Every bubble holding the same amount of air is the same bubble, which means that a finished bubble tells you nothing at all about how it was made. All of that got thrown away on the way down.

That throwing away is the whole reason it works. If a bubble kept some record of how it was blown, then bubbles blown differently would end up looking different, and there would be no single answer for the film to find. Forgetting is what lets many different beginnings arrive at one ending.

Hold onto that idea, because everything that follows is the same fall downhill, and the only thing that changes is how much survives it.

One bubble is easy; many is not

A single bubble has a simple job: wrap a volume in as little surface as possible. Things get harder when you have to divide a space into many cells that share their walls, which is what a foam does, and what bees do with wax.

On a flat surface, the answer is beautiful. Hexagons win. Thomas Hales proved in 1999 that no other way of cutting a flat sheet into equal-area cells uses less wall, including irregular ways and mixtures of different shapes. Circles would waste wax in the gaps between them, and squares need more wall for the same area, so hexagons it is.

In three dimensions, though, the neat answer loses. Lord Kelvin proposed a single repeating cell shape for dividing space into equal volumes, and it held

the record for over a century, until Weaire and Phelan found something better in 1993 using two different cell shapes together. So the moment you leave the flat page, the tidiest possible answer stops being the best one. Irregularity is not something living things brought into a tidy world. It was already there in the foam.

The honeycomb keeps one thing

Bees do not calculate hexagons. They build roughly round cells, and the wax, warmed by the colony, flows and settles until the walls meet at the same angles a soap film would choose. The comb is a **relaxation**, exactly like the bubble.

But something does survive this one. The wax argument says nothing about how big the cells should be; if saving wax were the only goal, bees would build fewer, bigger cells, since bigger cells use less wall for the honey they hold. Instead the width is fixed by the **larva** that has to fit inside. So one single fact about the occupant, its girth, is stamped into the finished comb, and nothing else about it is.

The protein keeps almost everything

A **protein** starts as a long chain of **amino acids** and folds itself into a three-dimensional shape. This is the same kind of downhill fall, driven by the water-hating parts of the chain hiding themselves on the inside. Nothing guides it. It simply settles.

Yet it does not settle into a sphere or a hexagon. It settles into one specific lumpy shape, different for every different chain, with a different job at almost every point on its surface. The order of the chain decides which shapes are low and which are high, so this time the fall does not wipe the information out. It develops it, the way a photograph develops.

That is why proteins are lumpy, and it is not a failure to be neat. When every part of a surface has a different job, being neat would mean doing those jobs badly. You can check this inside biology: the proteins that do the same thing along their whole length, like the **collagen** in your tendons and the **keratin** in your hair, really are regular and repeating. It is the ones that have to recognise particular molecules that are lumpy.

There is one more twist. A protein's shape is not a clean design for the job it does today, because **evolution** never starts from a blank chain. It edits a chain already in use. So some parts of the shape are there because they are needed now, some are left over from conditions that changed long ago, and some are there simply because removing them never helped. A leftover can even become essential later, once other parts are built around it. Yesterday's accident becomes today's requirement.

Astronomers have a version of the bubble's amnesia, by the way: two black holes with the same **mass**, **charge** and **spin** are impossible to tell apart from outside, no matter how different the stars that made them were. Smooth shapes are what you get when nothing about the past is allowed through.

Remembering is not the same as acting

It is tempting to say the protein is making decisions, since a pocket in its surface accepts one molecule and turns away others. Compare it to **Maxwell's demon**, an imaginary creature who watches individual particles and opens a door depending on what he sees. The comparison breaks on the watching. The pocket sees nothing and learns nothing. It is a shape with holes of the right size, and its choosiness was decided long ago by the sequence, not in the moment.

The real change comes with **molecular machines**. **Kinesin** walks along a track inside your cells, dragging cargo, and **ATP synthase** turns a flow of protons into chemical fuel. These do not just sit at the bottom of a fall. They spend energy every single cycle, and they stop the instant the supply stops.

Even that is not enough on its own, though, because a candle flame is fed energy continuously and holds a shape, and a flame passes nothing on. Energy supplies the effort; **inheritance** supplies the memory; **selection** decides what gets kept. That combination is why a machine built of protein could get better over millions of years and a flame could not.

The one sentence

The physics never changed. Everything here fell downhill and stopped. What changed is where the information was sitting: in the bubble it was in the starting shape, so the fall destroyed it; in the protein it was in the sequence, so the same fall revealed it; in the machine, the fall is harnessed to make something happen that would otherwise be too unlikely.

Medium, occupant, inheritance, power.

At the last step, a surface stops being an answer to a problem and starts being a machine for answering one.